KnowFlow Engineering Guide

Current-State Technical Reference

M1–M8 · KnowFlow V1

15 October 2026

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This guide describes the implementation currently in this repository. Code and migrations are the authority when an older plan differs. It is written to help you run, trace, debug, and explain KnowFlow, not to document every line of generated or boilerplate code.

> **V1 AI boundary:** KnowFlow does not use a generative LLM. Ask KnowFlow does not write, summarize, or paraphrase an answer. It retrieves and presents passages from authorized source documents. Local E5 embeddings support retrieval; they are not a chatbot and do not call a paid inference API.

1. KnowFlow at a glance

Organizations keep policies, procedures, handbooks, and technical guidance in separate files and folders. Finding the applicable passage takes time, and a search result is only useful if the person can verify its source.

KnowFlow provides a workspace for managing those documents and locating relevant, page-aware passages. A workspace can organize documents into **Knowledge Areas**, for example HR, Engineering, IT & Security, and Operations.

- **Owner:** manages Knowledge Areas, documents, and workspace membership; can also search and ask questions.
- **Member:** browses authorized areas and documents, downloads sources, searches, and asks questions. Members do not receive the Owner's management capabilities.
- **Ask KnowFlow:** takes one natural-language question and returns best matching evidence, optional supporting evidence, or clearly labeled potentially related passages. It is not a chat or conversation history.

A typical flow is: an Owner creates a workspace, creates a Knowledge Area, and uploads a text-based PDF. The durable worker extracts page text, chunks it, and creates local embeddings. When processing completes, Owners and Members can search or ask questions. The result identifies the document and page, and its protected source link opens the original document.

V1 deliberately excludes generative answers, OCR, chat history, DOCX/image ingestion, document version comparison, and analytics infrastructure. Weak evidence is not presented as an answer. The user inspects the source and interprets it.

KnowFlow in 30 seconds

> KnowFlow is a multi-tenant document knowledge platform. Teams organize private PDFs into workspaces and Knowledge Areas. A PostgreSQL-backed worker extracts page-aware passages and local E5 embeddings. PostgreSQL full-text search and pgvector find candidate passages, and Ask KnowFlow presents selected verbatim evidence with protected document/page sources. WorkOS authenticates people; KnowFlow workspace memberships authorize access. V1 has no generative LLM or paid AI inference dependency.

2. Current system architecture

The application is a **modular monolith**: one backend codebase contains identity, tenancy, document, processing, embedding, and search modules. Local development runs the API and worker as separate processes from the same backend image. The constrained hosted portfolio deployment runs the existing durable worker loop inside the FastAPI process and shares one ONNX embedding session with Ask/search. This changes the process topology, not the job model or domain architecture.

Component responsibilities

- **Browser:** displays the application and submits user actions. It receives an encrypted HttpOnly session cookie, not the backend bearer token.
- **Next.js frontend/BFF:** renders pages, performs server-side session handling, calls FastAPI, and streams authorized document content back to the browser. The BFF is a Backend-for-Frontend: a server-side layer that adapts the API for the web UI while keeping credentials out of browser JavaScript.
- **WorkOS AuthKit:** handles the identity flow, account verification, and sign-in session. It answers "who signed in?"; it does not assign KnowFlow workspace roles.
- **FastAPI:** validates API credentials, checks membership and roles, validates requests, orchestrates services, and reads/writes PostgreSQL and object storage.
- **PostgreSQL:** stores users, workspaces, memberships, areas, document metadata, processing jobs, extracted pages, chunks, and vectors. pgvector adds a vector column and similarity operators to PostgreSQL.

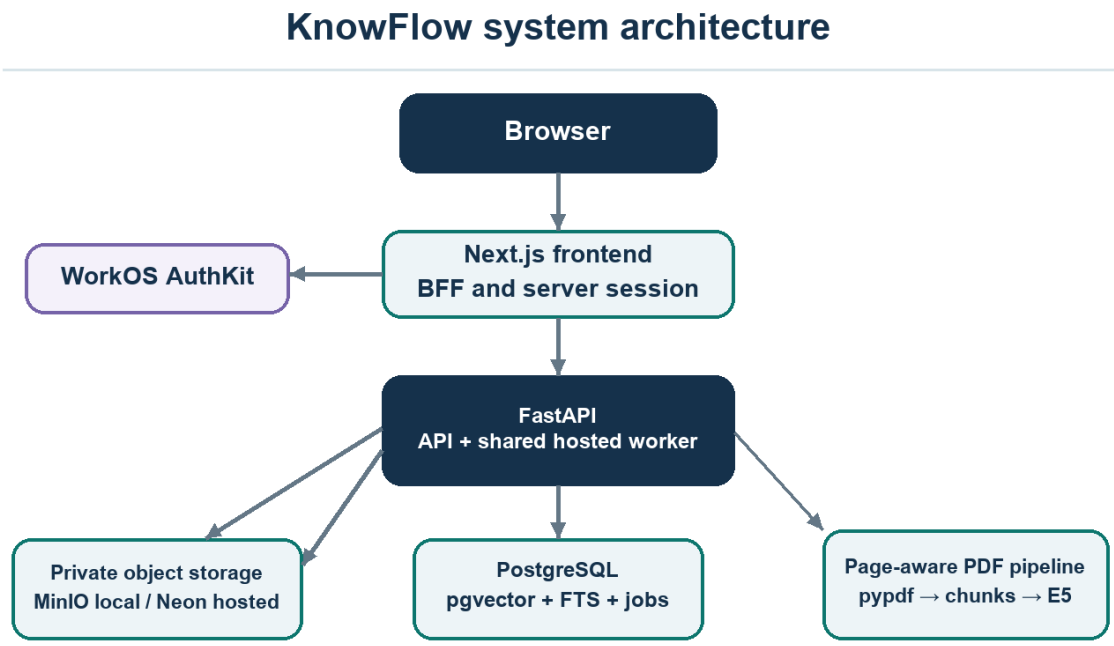


Figure 1: Overall system architecture

- **Object storage:** MinIO provides local S3-compatible private storage. The hosted demo uses a private Neon Object Storage bucket through the same S3 adapter. PostgreSQL stores metadata and an opaque storage key, not the PDF bytes.
- **Worker:** polls PostgreSQL, claims a job safely, downloads its PDF through the storage abstraction, extracts/chunks text, creates local embeddings, and commits derived output. It is a separate Compose service locally and an in-process thread in the hosted API.
- **E5 model:** a pretrained text embedding model run on CPU. Local development uses Sentence Transformers/PyTorch; the hosted backend uses CPU ONNX Runtime with the same pinned E5-small-v2 model and one shared session. It maps text to numeric vectors; it does not generate prose.
- **Retrieval and evidence layers:** M5 finds candidate passages with lexical and semantic search. M6 evaluates and groups evidence deterministically; it does not call an LLM.

3. Technology stack

Technology	Purpose in KnowFlow	Why it was chosen
Next.js	Web pages, server rendering, server actions, BFF routes	Keeps authenticated browser-to-API work on the server and supports a coherent full-stack app.
React	UI components and interactive controls	Provides the component model used by Next.js.
TypeScript	Frontend types and compile-time checks	Catches interface mistakes in application code before runtime.
FastAPI	REST API	Python API framework with request validation, dependency injection, and OpenAPI support.
Python	API, worker, processing, embeddings	Shares the document pipeline and model runtime in one backend codebase.
PostgreSQL	Relational source of truth and durable jobs	Supports transactions, constraints, FTS, and the expected modest V1 workload.

Technology	Purpose in KnowFlow	Why it was chosen
pgvector	vector(384) storage and cosine retrieval	Keeps vector search beside tenant and document records without another database.
SQLAlchemy	ORM and database access	Models the relational schema and supports explicit tenant-scoped queries.
Alembic	Schema migrations	Applies a versioned database change sequence from checked-in revisions.
MinIO / Neon private S3-compatible storage	Private original PDF storage	MinIO supports local development; the hosted private bucket fits the existing S3 adapter and keeps PDFs authorization-gated.
Docker / Compose	Local service environment	Starts web, API, worker, database, and object storage consistently.
WorkOS AuthKit	Managed user authentication	Avoids implementing password, verification, and recovery systems in the app.
pypdf	PDF parsing and page text extraction	Extracts selectable text and preserves page numbers; OCR is outside V1.
intfloat /e5−small−v2, Sentence Transformers, ONNX Runtime	Local text embeddings	One pinned 384-dimensional model supports PyTorch locally and a quantized CPU ONNX runtime within the hosted memory budget; no inference API is used.
Render Free	Public portfolio web hosting	Hosts the Next.js frontend and FastAPI/worker separately from the persistent data layer; accepts cold starts and low resource limits.
Neon	Hosted PostgreSQL, pgvector, FTS, durable jobs, private S3-compatible object storage	One hosted data provider serves the tiny portfolio workload while keeping the existing PostgreSQL and S3 interfaces.
GitHub Actions	Continuous integration	Runs backend and frontend quality gates on pull requests and main pushes; it does not deploy.

4. Repository structure

Current source paths (generated .next, node_modules, and caches are omitted):

```
.
|-- README.md
|-- docker-compose.yml
|-- render.yaml
|-- .env.example
|-- .github/workflows/ci.yml
|-- apps/web/
|   |-- Dockerfile
|   |-- src/proxy.ts
|   |-- src/lib/{api.ts,require-api-success.ts,ui-helpers.mjs}
|   |-- src/app/actions.ts
|   |-- src/app/api/health/{live,ready}/route.ts
|   |-- src/app/api/workspaces/[workspaceId]/
|   |   |-- ask/route.ts
|   |   |-- search/route.ts
|   |   +-- documents/[documentId]/content/route.ts
|   |-- src/app/components/{ConfirmSubmit,PendingSubmitButton,BackendLoadingScreen,Backend
```

```
|   +-- src/app/workspaces/
|       |-- page.tsx
|       +-- [workspaceId]/
|           |-- layout.tsx, WorkspaceNavigation.tsx, page.tsx
|           |-- search/{page.tsx,SearchForm.tsx}
|           |-- documents/{page.tsx,[documentId]/page.tsx}
|           |-- knowledge-areas/page.tsx
|           +-- members/page.tsx
|-- apps/web/tests/ui-helpers.test.mjs
|-- backend/
|   |-- Dockerfile, Dockerfile.onnx, pyproject.toml, alembic.ini
|   |-- app/
|   |   |-- main.py
|   |   |-- core/{config.py,errors.py}
|   |   |-- db/{models.py,session.py}
|   |   |-- identity/, tenancy/, knowledge_areas/, documents/
|   |   |-- jobs/, processing/, embeddings/, search/
|   |   +-- api/health.py
|   |-- migrations/
|   |   |-- env.py
|   |   +-- versions/
|   |       |-- 0001_enable_pgvector.py
|   |       |-- 0002_identity_and_tenancy.py
|   |       |-- 0003_knowledge_areas_and_documents.py
|   |       |-- 0004_processing_jobs_and_chunks.py
|   |       +-- 0005_embeddings_retrieval.py
|   |-- scripts/ (pinned ONNX artifact build and embedding benchmark/comparison tools)
|   +-- tests/ (API, processing, embedding, retrieval, PostgreSQL)
+-- docs/
    |-- KNOWFLOW_ENGINEERING_GUIDE.md / .pdf
    |-- KnowFlow_Project_Blueprint.pdf (earlier planning reference)
    |-- KnowFlow_V1_Detailed_Design.pdf (earlier design reference)
    |-- adr/0001...0011
    +-- samples/ (development/test PDFs)
```

- apps/web/src/app/workspaces/[workspaceId]/layout.tsx is the shared authenticated workspace shell. Its child pages share the top bar and workspace navigation.
- apps/web/src/app/actions.ts contains server actions for form operations. Server actions call the server-only API helper rather than making the browser responsible for bearer credentials.
- apps/web/src/app/api/... contains Next.js BFF routes for Ask, search, protected PDF streaming, and backend readiness. BackendStartupNotice and BackendLoadingScreen provide a bounded cold-start/wait experience.
- apps/web/src/lib/api.ts centralizes server-side API calls; proxy.ts, AuthKit routes, and callback implement the login/session boundary.
- backend/app/<domain>/ groups API routes and services by concern. main.py mounts the routers.
- backend/app/db/models.py is the current SQLAlchemy model definition; db/session.py creates the engine and request session dependency.
- backend/app/jobs/ claims and runs durable document-processing work. processing/ contains parsing, normalization, chunking, and atomic persistence. embeddings/ isolates the local model provider. search/ implements M5 retrieval and M6 evidence selection.
- backend/migrations/versions/ contains the actual schema history through revision 0005.
- backend/tests/ separates API, job, processing, embedding, evidence, and PostgreSQL-specific coverage. SQLite fixtures are not a substitute for pgvector/FTS integration tests.
- docs/adr/ records accepted design decisions. docs/samples/ are fixtures and are not special-cased in production logic.

5. Database architecture

SQLAlchemy models are in backend/app/db/models.py; migrations in backend/migrations/versions/ define what PostgreSQL has actually been instructed to create.

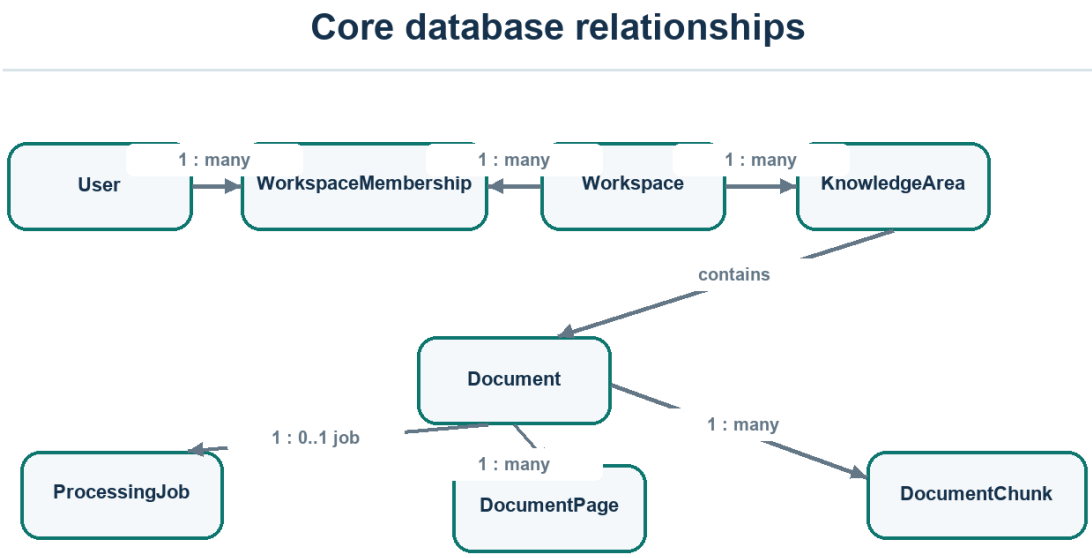


Figure 2: Database entity relationships

Model / table	Purpose and important fields
User / users	Internal user ID (UUID), immutable WorkOS auth_subject, normalized email, verification state, display name, timestamps. auth_subject and email are unique.
Workspace / workspaces	Tenant root: UUID, name, unique slug, creator, timestamps.
WorkspaceMembership / workspace_memberships	Join table for users and workspaces. Composite primary key (workspace_id, user_id), role OWNER or MEMBER, creator and timestamps. It is the authorization authority.
KnowledgeArea / knowledge_areas	Workspace organization unit with name, normalized name, optional description, creator, timestamps. (workspace_id, normalized_name) is unique.
Document / documents	PDF metadata, workspace and area, uploader, original filename, opaque storage_key, content type, byte/page counts, SHA-256, lifecycle and deletion/processing fields. The actual PDF bytes are stored elsewhere.
ProcessingJob / processing_jobs	One durable PROCESS_DOCUMENT job per workspace/document. Includes state, attempt count, max attempts, availability, lease timestamp/owner/token, safe error code/message, and completion time.
DocumentPage / document_pages	Extracted normalized text and hash for each 1-based source page. (document_id, page_number) is unique.
DocumentChunk / document_chunks	Ordered text chunk, content hash, inclusive page range, nullable token count, vector(384), model/version/dimension metadata, and embedding timestamp. The FTS expression has a GIN index.

IDs, relationships, and integrity

UUIDs provide opaque identifiers that are convenient across API boundaries; they are not access control. Tenant-owned records include workspace_id. Composite foreign keys, for example (workspace_id, document_id) from a chunk to a document, make it impossible to attach a chunk from one workspace to a document in another workspace even if the IDs are valid individually.

The DocumentChunk vector is nullable in the schema because rows may be constructed before embedding or during controlled migration states. The check constraint requires embedding model, version, fixed dimension 384,

and timestamp to be present together with a vector. Current normal worker success persists fully embedded chunks; retrieval further filters to the configured model identity/version/dimensions.

Authorization versus integrity

A database constraint answers "are these rows structurally consistent?" It does not answer "may this user read them?" FastAPI first checks internal membership and then uses workspace-scoped SQL. Composite constraints provide a second line of defense against invalid relationships; they do not replace authorization.

Original PDFs are in object storage because the database is better suited to relational metadata, transactions, and derived searchable text than large binary payloads. The database stores an opaque storage key and digest. MinIO URLs and raw vector values are not sent to users.

6. Authentication and authorization

Authentication establishes who the caller is. **Authorization** decides what that user can do in a specific workspace.

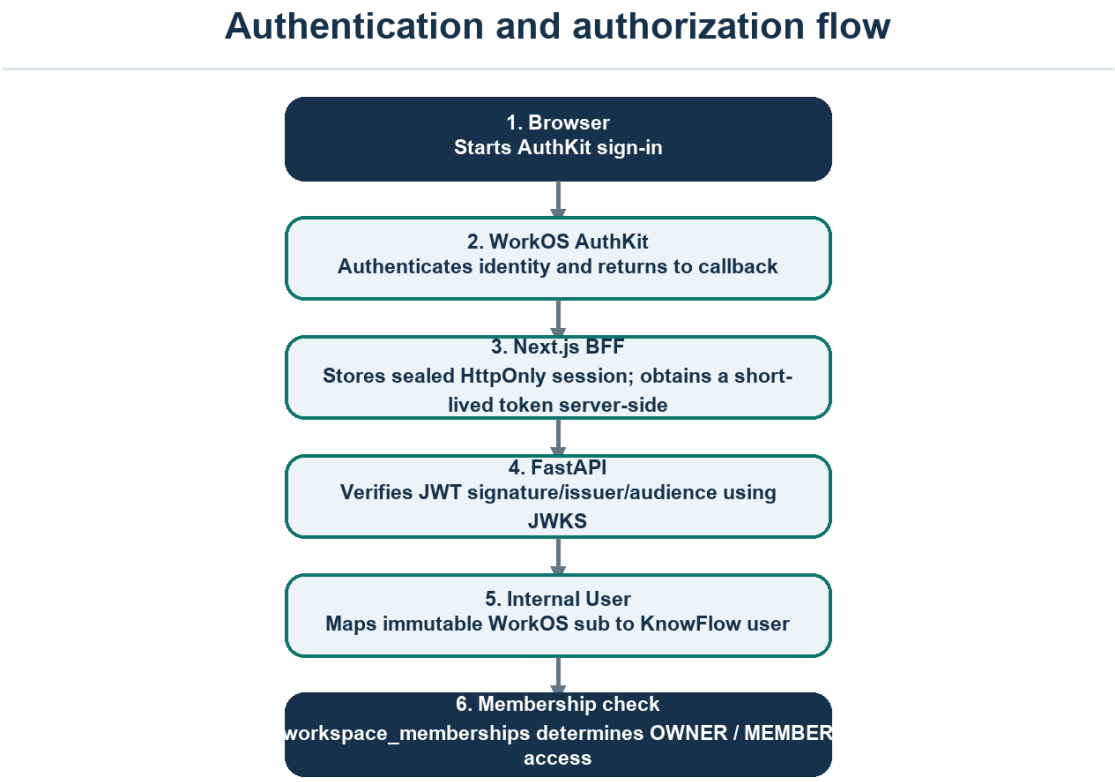


Figure 3: Authentication request flow

The actual flow is implemented across apps/web/src/proxy.ts, apps/web/src/app/sign−in/route.ts, callback/route.ts, apps/web/src/lib/api.ts, backend/app/identity/workos.py, identity/dependencies.py, and identity/service.py.

1. AuthKit performs the OIDC authorization-code flow with PKCE. PKCE binds the callback to the sign-in request and helps prevent code interception/replay.
2. Next.js stores the session in a sealed HttpOnly cookie. HttpOnly prevents normal browser JavaScript from reading it. Server components/actions obtain a short-lived access token and make a server-to-server FastAPI request.
3. FastAPI checks the RS256 JWT using WorkOS JWKS (public signing keys), expected issuer, configured audience, expiry, and required subject. An audience is the API identifier the token is intended for.
4. The immutable WorkOS sub is mapped to User.auth_subject; mutable email is profile data, not the identity key.
5. require_workspace_access() checks (user_id, workspace_id) in workspace_memberships. require_owner() gates Owner actions. The backend is authoritative even if a UI control is hidden.

WorkOS Organizations/Roles are not used for KnowFlow authorization. Workspace creation gives the creator the initial Owner membership. Membership provisioning accepts an already registered, verified WorkOS user; V1 has

no email invitation workflow.

7. Multi-tenancy and security

- **Membership first:** backend/app/tenancy/policies.py loads the workspace only through a matching membership. A non-member receives 404, which avoids disclosing whether another tenant's workspace exists.
- **Role checks:** Owners manage areas, documents, and membership. Members can read the permitted workspace content and ask/search. Some Owner-only actions return 403 when the workspace itself is accessible.
- **Workspace-scoped queries:** service methods carry WorkspaceAccess and constrain records by that workspace. Search filters the workspace and only active READY documents with current embeddings before returning candidate text.
- **Composite integrity:** composite foreign keys tie documents to an area and jobs/pages/chunks to the same workspace/document pair.
- **No PostgreSQL RLS in V1:** application-level membership checks and scoped SQL are explicit and tested. RLS would add a second policy/configuration system and session context requirement. It remains a possible defense-in-depth measure, not a current feature.
- **Private source access:** object storage is not public. The browser accesses the original through a Next.js route that forwards a short-lived token; FastAPI checks membership and document ownership before streaming. Knowing a document UUID or source path does not grant access.
- **Safe exposure:** response schemas omit vectors. The UI sees filename, text, page, and protected app route, not storage keys or object-store URLs. Request errors are mapped to safe messages; diagnostic details remain server side.

These mechanisms are covered by backend/tests/test_tenancy_api.py, test_documents_api.py, test_search.py, and test_postgres_retrieval.py among others. Tests verify foreign workspace IDs do not expose documents, chunks, or retrieval results.

8. Document lifecycle

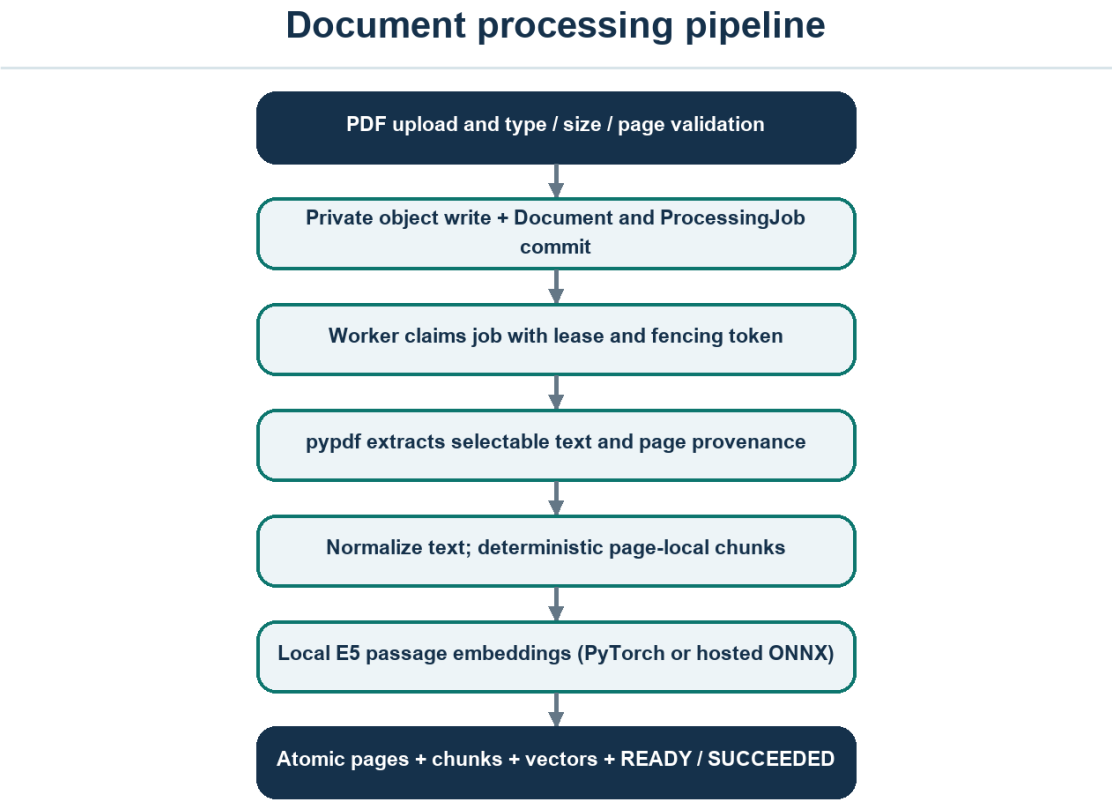


Figure 4: Document processing pipeline

Upload path

apps/web/src/app/workspaces/[workspaceId]/documents/page.tsx submits an Owner upload using a server action in apps/web/src/app/actions.ts. The Next.js server calls FastAPI using apiFetch. backend/app/documents/router.py checks access and delegates to documents/service.py; documents/validation.py enforces a PDF structure and limits. Local defaults are 20 MiB and 250 pages; the hosted portfolio backend sets an 8 MiB upload limit and 250 pages to fit the Next.js action boundary and free-tier resources. Password-protected PDFs are unsupported. The stored metadata records application/pdf, byte size, page count, SHA-256, original filename, and an opaque key.

The service writes the private object using ObjectStorage (documents/storage.py) and then commits document metadata and the PROCESS_DOCUMENT job together. If the database commit fails after a successful object write, the service attempts a compensating object delete. Cross-system storage and database writes cannot be one atomic transaction, so compensation and retryable cleanup handle partial failure.

Lifecycle states

- UPLOADED: accepted and waiting for worker processing or reprocessing.
- PROCESSING: a worker has claimed the job or a retry is pending.
- READY: extracted pages, chunks, and valid current-model vectors were committed.
- FAILED: processing exhausted retries or hit a permanent input/configuration failure.
- DELETING: access has been revoked and physical cleanup/tombstone handling is underway.

The API returns document status and safe processing error information. The frontend shows Uploaded, Processing, Ready, or Failed and explains that a document is searchable only once Ready.

9. Worker and PostgreSQL job queue

The queue is a row in processing_jobs, not a separate Redis/Celery/RabbitMQ service. PostgreSQL is already required, the portfolio workload is modest, and a committed row survives process restarts.

Claim and concurrency

backend/app/jobs/service.py:lock_next_eligible_job() selects eligible work in stable order and issues SELECT ... FOR UPDATE SKIP LOCKED LIMIT 1. A row lock protects a selected job while it is claimed. SKIP LOCKED lets a second worker skip that row and claim another one instead of waiting for or duplicating the same job. PostgreSQL is required for this behavior; SQLite tests cannot prove it.

A successful claim increments attempt_count, sets RUNNING, stores locked_at and locked_by, assigns a random lease_token, and marks the document PROCESSING. The token is a **fencing token**: every heartbeat/final write must present the current token. If an old worker resumes after its lease has expired and another worker has claimed the job, its stale token prevents its result from overwriting the newer attempt.

Heartbeat, timeout, retries

Local polling defaults to every 2 seconds (KNOWFLOW_WORKER_POLL_INTERVAL_SECONDS). Render config sets a 600-second idle poll interval to avoid keeping Neon active unnecessarily; successful uploads wake the in-process worker immediately. The hosted backend processes one document at a time and forces embedding batch size 1. The configured lease is 90 seconds and heartbeat 30 seconds; the processing timeout is separately configurable. A heartbeat refreshes locked_at. If the process crashes, the lease eventually expires and the job becomes claimable again. If the worker is deliberately shut down, the process exits cleanly; uncompleted work is recovered through lease expiry.

record_failure() distinguishes retryable errors (for example temporary storage/model transport/runtime errors) from permanent document/configuration errors (for example malformed/encrypted PDF or incompatible vector output). Retry delay defaults are 30, 120, and 600 seconds with up to four attempts. The same job row records the current attempt count and last safe error; V1 does not store an immutable row for every attempt.

Idempotency and atomic completion

Idempotency means repeating work produces the same intended state rather than duplicate/corrupt output. persist_processed_output() verifies job ID, workspace/document IDs, RUNNING state, and fencing token. Inside one database transaction it deletes prior page/chunk rows for that document, inserts the new pages/chunks, marks the document READY, and the job SUCCEEDED. If the transaction fails, none of those database changes commit. If the source was deleted, it cancels work and does not restore access or write chunks.

The worker logic is in backend/app/jobs/worker.py, with claim/retry behavior in jobs/service.py and transactional output in processing/persistence.py.

10. PDF extraction and chunking

KnowFlow accepts text-based/selectable PDFs only. backend/app/processing/extraction.py:extract_pdf_pages() retrieves an object through the storage abstraction, verifies byte length and SHA-256 against upload metadata, then uses pypdf.PdfReader to extract pages from 1 onward. Encrypted, malformed, missing, integrity-mismatched, and no-meaningful-text PDFs produce explicit safe processing errors. OCR is not performed.

normalize_page_text() applies Unicode NFKC normalization, removes null/soft-hyphen characters, standardizes line endings and collapses whitespace while retaining useful paragraph boundaries. Every page has a stable content hash and its original 1-based page number.

backend/app/processing/chunking.py:chunk_pages() processes pages in page order. Defaults are **1,800 characters per chunk and 240 characters of overlap** (KNOWFLOW_CHUNK_SIZE_CHARS, KNOWFLOW_CHUNK_OVERLAP_CHARS). It prefers a paragraph break, then a sentence break, then whitespace, otherwise a hard boundary. It never crosses page boundaries in the current implementation. Chunk indices are zero-based within the document; each chunk stores page_start and page_end (equal for current page-local chunks) and a content hash.

Page-aware chunks preserve the link from a search hit back to a PDF page. Overlap reduces the chance that a relevant sentence at a boundary loses its nearby context. Character-based splitting is deterministic and simple; it is not a semantic paragraph parser, and the embedding model truncates each input beyond its 512-token context.

11. Local embeddings

An **embedding** is a list of numbers that represents text in a space where texts with related meaning often have nearby vectors. It supports similarity search; it does not produce an answer or new text.

KnowFlow uses pretrained **intfloat/e5−small−v2**, pinned to revision ffb93f3bd4047442299a41ebb6fa998a38507c52. It is MIT-licensed, has 384 output dimensions and supports up to 512 tokens. Inference uses CPU and does not require an API key, GPU, paid service, or model training.

Local development defaults to Sentence Transformers/PyTorch. API and worker are separate local processes and each may load its own model instance. The hosted Render image uses a portable INT8 ONNX artifact (dynamic quantization; it does not rely on AVX512/VNNI), CPU ONNX Runtime, and no PyTorch or Sentence Transformers packages. The model artifact is baked into the backend image so runtime model download is not required. At the enforced batch size of 1, the local Docker 512 MiB proof measured 306.16 MiB process peak RSS (261.64 MiB cgroup peak), no OOM, and about 206 MiB process-RSS headroom. The hosted process shares one model session between query and document embedding. This is a constrained portfolio configuration, not a general large-corpus throughput target.

The measured ONNX model artifact was about **66.54 MiB**. Median query embedding latency in the benchmark was about **19 ms**; embedding 80 sample chunks took about **27.7 seconds** at batch size 1. A 36-vector PyTorch/ONNX comparison had paired cosine similarity from approximately **0.994 to 0.996**, and the vacation-time/annual-leave retrieval case retained the correct top result. These measurements support the portfolio workload but are not a throughput guarantee for larger documents or different hardware.

The provider contract is backend/app/embeddings/provider.py; sentence_transformers.py validates the model identity, vector length, and finite values. Document text gets the prefix passage:; search/Ask questions get query:. This is the E5 retrieval convention. Embeddings are normalized and then stored directly on document_chunks.embedding as vector(384). The row also stores model ID plus immutable revision, pipeline embedding_version, dimensions, and embedded_at. The PyTorch pipeline and ONNX INT8 pipeline use distinct embedding-version metadata. Paired-vector comparisons measured cosine similarity around 0.994–0.996 and retrieval regressions passed, but the versions are not treated as interchangeable: do not mix vectors; reprocess active documents when changing provider/version.

The Compose model_cache volume is mounted at /models and shared by API and worker. The first successful local model load downloads pinned files from Hugging Face, so initial setup requires internet. The cache completion marker makes later model loads local-only; ordinary docker compose down preserves it. docker compose down −v removes it. Weights are not committed to Git. Docker/Linux install CPU-only PyTorch; macOS inference is also configured to CPU. The hosted image includes its pinned ONNX artifact at build time and does not depend on ephemeral filesystem persistence or a runtime download.

Example: the user asks, “How much vacation time do staff get?” The stored policy says, “Full-time employees receive 25 working days of paid annual leave per calendar year.” FTS may have little exact token overlap between “vacation” and “annual leave.” E5 can place the two texts near each other because it was trained to represent related meaning. The retrieved source remains the actual quote; KnowFlow does not rewrite it as an answer.

12. Hybrid retrieval

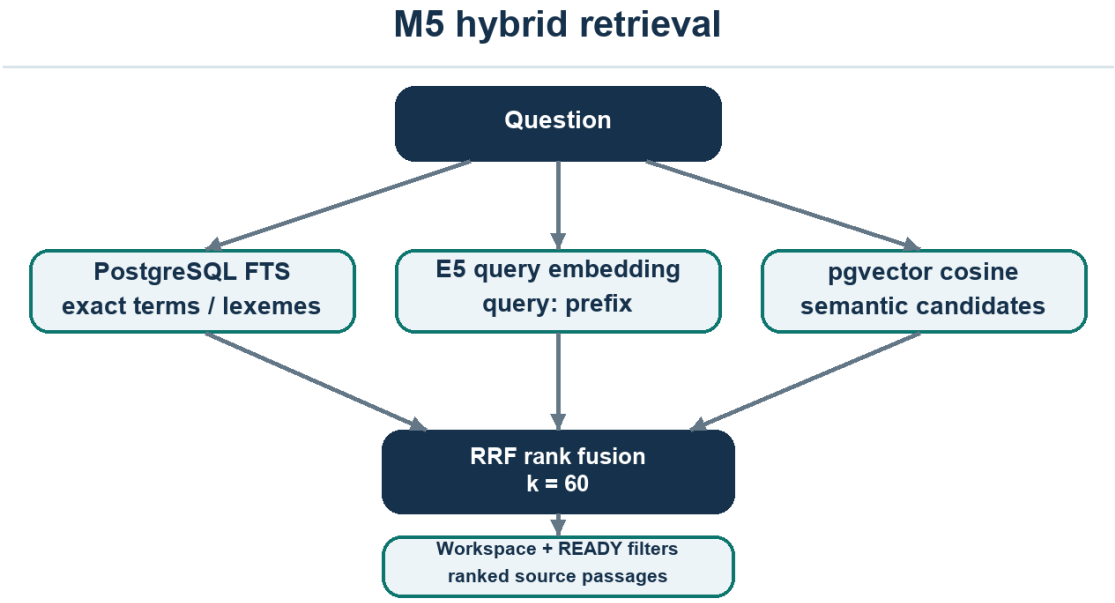


Figure 5: M5 hybrid retrieval pipeline

PostgreSQL full-text search (FTS) converts text to language-aware lexemes and matches a web-style query. The search service uses websearch_to_tsquery('english', ...), to_tsvector('english', chunk.text), ts_rank_cd, and a GIN expression index. FTS is good at exact words and policy terms, but misses synonyms and paraphrases.

Vector search embeds the question locally and orders stored vectors by pgvector cosine distance. Cosine similarity compares vector direction; normalized vectors allow the service to derive similarity as 1 − cosine_distance. It is useful for paraphrases, but a semantically related passage can still be the wrong policy or only a test/reference question.

backend/app/search/service.py:retrieve() performs the two candidate queries independently and requires the requested workspace, an active non-deleted READY document, and vector metadata matching the configured model/version/dimension. It validates an optional Knowledge Area belongs to that workspace; an inaccessible area ID returns 404.

The defaults are 30 FTS candidates, 30 vector candidates, 6 final results, and **RRF k=60**, all configurable. Reciprocal Rank Fusion uses 1 / (k + rank) for each list; it combines rank positions instead of adding raw FTS scores to cosine similarities, whose scales are different. A chunk appearing in both lists accumulates both contributions. Deterministic IDs/indexes break ties. RRF ranks are not calibrated probabilities.

The REST response includes chunk/document IDs, filename, text, page range, rank, RRF score, and optional component ranks. Internal keyword score and cosine similarity fields are excluded from the public M5 schema. No vector is exposed. POST /api/v1/workspaces/{workspace_id}/search is available to both workspace roles.

13. Ask KnowFlow: M6 evidence, not generation

POST /api/v1/workspaces/{workspace_id}/ask calls the same M5 retrieval helper as /search; it does not duplicate or change the retrieval algorithm. It then runs backend/app/search/evidence.py:select_evidence().

What the evidence layer does

1. **Passage extraction:** extract_passage() splits retrieved chunk text into sentences and scores sentence overlap, answer-type cues, declarative structure, question-like structure, and reference markers. It chooses

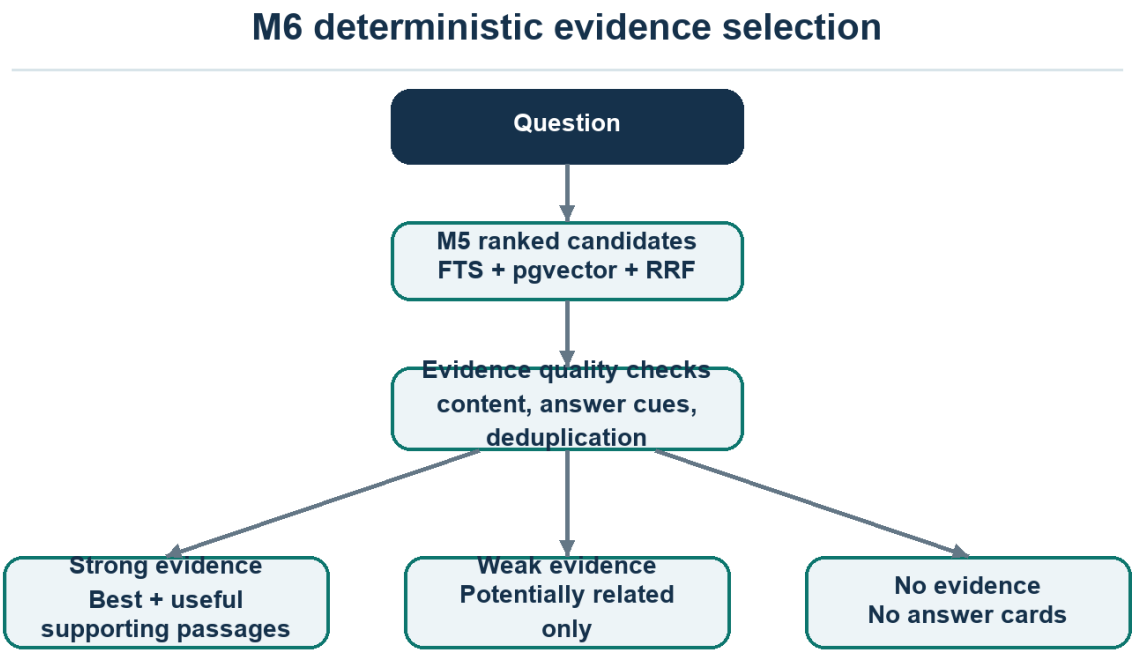


Figure 6: M6 evidence-selection states

source sentences and short adjacent context when useful. It only normalizes whitespace; it never paraphrases. If a safe smaller passage cannot be isolated, the original chunk is used.

2. **Quality assessment:** the selector evaluates substantive declarative content and checks for question/reference material. Query terms are normalized with lightweight morphology and generic words removed. Lexical support combines FTS presence, meaningful term overlap, and distinguishing-concept coverage. For semantic support it uses the actual candidate cosine similarity from M5, not its rank or RRF score. The configured minimum vector similarity is 0.72, but it only contributes with substantive evidence and a compatible answer-type cue (frequency, quantity, timing, or responsible role). These are heuristics, not a probability.
3. **Deduplication:** normalized extracted text is compared across all candidates; near duplicates use token-set Jaccard similarity with default threshold 0.72. Same-document/same-page overlapping chunks also collapse. The best passage cannot reappear as supporting evidence. Supporting passages must add context; the default three is a maximum, not a quota.
4. **Evidence state:** evidence_found may return best and supporting evidence. weak_evidence has no best/-supporting cards and may return only related_evidence with an explicit weak label. no_evidence returns no evidence cards. The normal UI hides engineering rank and score details.
5. **Provenance:** every evidence item carries document ID/title, page start/end, and an application source_path. The frontend route opens a protected document page and adds a PDF page fragment. FastAPI checks membership/document scope again when streaming the original.

When separate policies disagree, KnowFlow presents each source passage with its own provenance. It does not synthesize or reconcile a single rule. This avoids pretending that one document overrides another. A deterministic evidence selector is not general natural-language reasoning: ambiguous or unusual questions may be missed or classified weakly. Semantic similarity by itself is insufficient evidence that a passage answers the requested concept.

14. Frontend and user experience

The shared workspace layout in apps/web/src/app/workspaces/[workspaceId]/layout.tsx fetches the authorized workspace on the server and renders the KnowFlow top bar, role, workspace switch, sign-out form, and navigation.

- **Overview:** [workspaceId]/page.tsx; summarizes existing area/document/status information and gives a path to Ask. It is not an analytics service.
- **Ask KnowFlow:** [workspaceId]/search/page.tsx and SearchForm.tsx; one question at a time, explicit loading/error/empty/evidence states, source pages, no chat bubbles or persisted conversation.
- **Knowledge Areas:** [workspaceId]/knowledge−areas/page.tsx; browsing for Members and management controls for Owners.
- **Documents:** [workspaceId]/documents/page.tsx; upload and library, status, protected source actions;

Owner management controls. The document detail page includes the protected PDF viewer.

- **Members:** [workspaceId]/members/page.tsx; Owner management controls, subject to final Owner protections in the backend.
- **Navigation:** desktop uses the left sidebar; mobile/tablet hides that sidebar and shows a header hamburger menu. The menu has all five destinations, active-page indication, aria−expanded, aria−controls, Escape handling, and closes after navigation. Desktop and mobile use the same destination list.

Next.js rendering and BFF boundaries

Server Components are rendered on the server by default and are used for authenticated data loading. Client Components are limited to interactions that need browser state, such as the Ask form or mobile menu. Server Actions handle form submissions. BFF route handlers in apps/web/src/app/api/... adapt Ask/search requests and stream protected document content. They call apiFetch with a server-held token; they do not pass that token to client JavaScript. FastAPI remains the API/security boundary and independently authorizes each request.

M7 adds visible pending states (PendingSubmitButton), explicit confirmation (ConfirmSubmit), accessible labels and focus indicators, textual document status, safe generic error UI, and responsive layouts. It does not add an alternate client data store or change backend policy. The frontend helper suite is in apps/web/tests/ui−helpers.test.mjs; backend tests remain the authority for security behavior.

15. Document deletion

Deletion separates **logical revocation** from **physical cleanup**. PostgreSQL cannot commit a transaction that atomically includes an object-store delete, so the app first commits a DELETING tombstone and deleted_at. It cancels unfinished processing jobs and deletes pages/chunks immediately. Workspace-scoped API reads, downloads, search, and worker selection exclude the deleted document, so access is revoked before storage availability is known.

The service then deletes the original object through the adapter. If storage is temporarily unavailable, the tombstone records a safe cleanup error and remains inaccessible; a repeated Owner delete operation can retry for up to the configured 24-hour cleanup window. After physical cleanup succeeds, cleanup_completed_at is set and purge_after is 7 days later. purge_expired_tombstones() removes eligible metadata opportunistically during document/area operations; there is no separate scheduled cleanup worker. After the retry window expires without cleanup, the API returns a safe operator-attention error while preserving the tombstone.

16. Local development and hosted deployment

Compose services and named volumes are defined in docker−compose.yml.

Service	Role	Persistent data / ports
web	Next.js development server	web_node_modules, web_next_cache; port 3000
api	FastAPI development server	mounts backend source and shared model_cache; port 8000
worker	Same backend image, separate process	mounts backend source and shared model_cache; no public port
db	PostgreSQL 16 with pgvector	postgres_data; port 5432
object−store	MinIO S3-compatible API and console	object_store_data; ports 9000 and 9001

The model cache volume persists independently. HF_HOME and SENTENCE_TRANSFORMERS_HOME point at /models. Local Compose remains the full developer topology: separate API and worker, local PostgreSQL/pgvector, MinIO, and default PyTorch embeddings.

```
docker compose up --build -d
docker compose run --rm api alembic upgrade head
docker compose logs -f worker
docker compose down
```

- up −−build −d builds images, creates the network/volumes if needed, starts services in dependency order, and returns without attaching to logs. Database readiness gates API/worker startup; MinIO is a started dependency.
- run −−rm api alembic upgrade head starts a temporary API-image container, connects it to Compose services, applies all unapplied migrations to the latest revision, then removes that one-off container. It does not generate migrations.
- logs −f worker follows the worker's polling and processing logs.
- down stops/removes containers and the Compose network but preserves named volumes and local database, object-store, dependency, and model-cache data.

  docker compose down −v removes the named volumes too. It deletes the local PostgreSQL database, uploaded MinIO files, and downloaded model cache (as well as web dependency/cache volumes). Use it only when intentionally resetting local data.

Copy .env.example to .env, configure WorkOS AuthKit callback/redirect values and audience, then start services. Local default database/object-store credentials are development-only. Apply migrations before readiness-dependent flows. On first model initialization, internet is needed to fetch the pinned E5 files; after the completion marker is written, the shared cache supports local-only loading.

Hosted portfolio topology

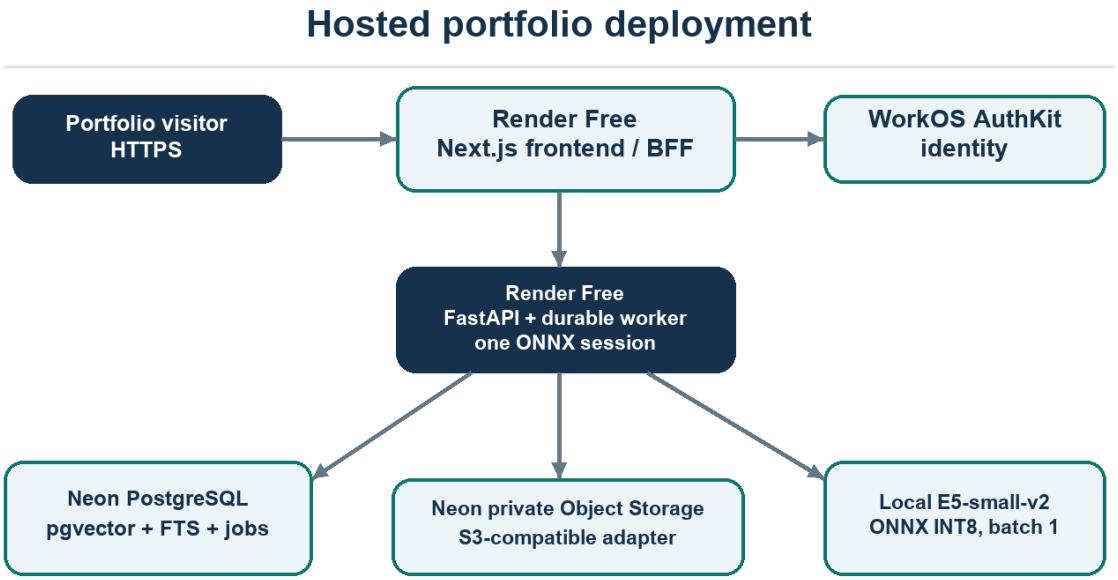


Figure 7: Hosted portfolio deployment

Render hosts two Free web services: knowflow−web and knowflow−backend; Neon is the database and private S3-compatible object store. There is no Render database and no public PDF bucket. WorkOS authenticates; KnowFlow memberships authorize. The backend receives database, storage, and WorkOS API credentials; the Next.js service receives its server-side WorkOS configuration and backend URL. Browser-visible configuration is limited to the redirect URL and never contains backend credentials.

The backend runs uvicorn app.main:app from backend/Dockerfile.onnx. The image contains CPU ONNX Runtime and the pinned INT8 E5 artifact, and excludes PyTorch and Sentence Transformers. Configure backend runtime values through render.yaml and Render's secret environment settings. Set these backend environment-variable names in the service dashboard:

KNOWFLOW_DATABASE_URL
KNOWFLOW_OBJECT_STORAGE_ENDPOINT_URL
KNOWFLOW_OBJECT_STORAGE_REGION
KNOWFLOW_OBJECT_STORAGE_BUCKET
KNOWFLOW_OBJECT_STORAGE_ACCESS_KEY
KNOWFLOW_OBJECT_STORAGE_SECRET_KEY
WORKOS_CLIENT_ID

WORKOS_API_KEY
WORKOS_ISSUER
WORKOS_TOKEN_AUDIENCE

The web service uses these environment-variable names:

WORKOS_CLIENT_ID
WORKOS_API_KEY
WORKOS_COOKIE_PASSWORD
API_INTERNAL_BASE_URL
NEXT_PUBLIC_WORKOS_REDIRECT_URI

Values belong in Render's service dashboard; never put secrets in Git or browser-exposed variables.

For the WorkOS production application, set the redirect/callback URI to <FRONTEND_URL>/callback, the initiate-login URI to <FRONTEND_URL>/sign−in, and the sign-out URI to <FRONTEND_URL>/. Use the same callback for NEXT_PUBLIC_WORKOS_REDIRECT_URI; configure the backend issuer and token audience from that same WorkOS environment. Keep local development redirect URIs alongside the production ones. The API key and cookie password stay only in server-side service settings.

The backend's non-secret runtime settings are:

KNOWFLOW_APP_ENV=production
KNOWFLOW_RUN_WORKER_IN_API=true
KNOWFLOW_EMBEDDING_BACKEND=onnx
KNOWFLOW_EMBEDDING_BATCH_SIZE=1
KNOWFLOW_WORKER_POLL_INTERVAL_SECONDS=600
KNOWFLOW_MAX_UPLOAD_BYTES=8388608
KNOWFLOW_MAX_PDF_PAGES=250

Render Free can stop services after inactivity. The web startup notice polls backend readiness for at most 24 attempts, with an 8-second request timeout and 5-second intervals, then offers Retry; this is bounded and sends no keep-alive traffic. When the backend starts, it initializes the single shared ONNX provider before readiness, starts the durable worker thread, then serves requests. An upload signals the worker immediately. The worker processes one job at a time. If the instance stops during a job, the job row remains in Neon and lease expiry makes it recoverable after restart.

Migrations and routine maintenance

For a release that changes the schema, run alembic upgrade head against the Neon URL using a one-off backend environment with the production secrets, then verify alembic current and alembic check. The current deployed head is 0005_embeddings_retrieval; the existing migrations were verified against Neon. Do not create a Render database. Keep Render auto-deploy disabled as configured: deploy the reviewed commit from the Render dashboard, check backend /health/live and /health/ready, then check frontend /api/health/live and /api/health/ready. For code/configuration changes use Render's manual deploy/restart controls; never edit production credentials in source control. Readiness checks the database, vector extension, initialized embedding provider, and in-process worker without embedding a health-check question.

Release smoke test

1. Open the public frontend; allow its bounded "Starting demo environment…" wait if the backend slept.
2. Sign in through WorkOS and verify the workspace loads.
3. Upload the sample HR text PDF and wait for READY.
4. Ask "How much vacation time do staff get?" and verify evidence for 25 working days with page 2 provenance.
5. Open the protected source and download it; verify unauthorized direct object access remains denied.
6. Delete the test document, confirm it disappears from normal access, then sign out.

Free-tier service sleep, limited memory/CPU/storage, low quotas, and no SLA are accepted portfolio trade-offs. Neon and Render quotas and account terms can change; confirm current dashboard limits before relying on the demo. No automated deployment workflow exists; GitHub Actions is CI only.

17. Database migrations

Alembic is configured by backend/alembic.ini and backend/migrations/env.py. The environment imports Base.metadata, reads the configured database URL, and uses the ordered revision / down_revision links below.

Alembic records the current revision in alembic_version.

Revision	Current migration purpose
0001_enable_pgvector	Enables the PostgreSQL vector extension.
0002_identity_and_tenancy	Creates users, workspaces, and Owner/Member memberships.
0003_areas_documents	Adds Knowledge Areas, document metadata/tombstones and workspace constraints/indexes.
0004_processing_jobs_chunks	Adds processing error fields, durable jobs, extracted pages, chunks, indexes, and enqueues existing eligible M3 documents.
0005_embeddings_retrieval	Adds vector(384), model/version/dimension/time metadata, embedding consistency constraint, FTS GIN index, and requeues eligible documents so the worker creates vectors.

alembic upgrade head applies unapplied revisions in chain order. alembic check compares current SQLAlchemy metadata against the connected PostgreSQL schema and reports model/schema drift; it does not repair drift or write a migration. Migration 0005 does not run embeddings inside DDL; the normal worker path reprocesses active documents.

18. Testing

Backend tests

Test file	Main concern
test_config.py	Runtime configuration and production safety invariants.
test_workos_auth.py	JWT/JWKS validation and internal user mapping.
test_tenancy_api.py	Membership roles, workspace access, and member management.
test_knowledge_areas_api.py	Area authorization, validation, and tenant scope.
test_documents_api.py	PDF validation, object storage, listing, access, deletion, retries.
test_jobs.py, test_postgres_job_concurrency.py	Claiming, retry state, lease/concurrency behavior.
test_processing.py	Page extraction, chunking, worker lifecycle, failures, persistence.
test_embeddings.py	Local provider contract, prefixes, dimensions, errors and metadata.
test_search.py, test_evidence.py	RRF/API tenant behavior and deterministic evidence selection.
test_postgres_retrieval.py	Real PostgreSQL pgvector/FTS behavior, candidate ranks, provenance and filters.

Most API/service tests use an isolated SQLite fixture and an auth dependency override. That proves application rules and general relational behavior, but SQLite does not implement PostgreSQL FTS, pgvector operators, PostgreSQL migration DDL, or SKIP LOCKED concurrency semantics. Those require a real PostgreSQL instance with pgvector and migrations. Model-provider unit tests use a fake provider/model to avoid downloads; sample PDF end-to-end retrieval requires the actual local model and Compose services.

Frontend and commands

The frontend has a small dependency-free Node test suite (apps/web/tests/ui−helpers.test.mjs) for role-capability helpers, document status labels, safe Ask errors, and navigation/menu contracts. It is not a browser automation suite.

```
# Compose backend quality checks
docker compose run —rm api ruff check app tests migrations
```

```
docker compose run --rm api pytest
docker compose run --rm api pytest tests/test_postgres_retrieval.py tests/test_postgres_job
docker compose run --rm api alembic check

# Frontend checks (or run in the web container)
cd apps/web
npm test
npm run lint
npm run typecheck
npm run build
```

The PostgreSQL integration tests need KNOWFLOW_DATABASE_URL to point to the Compose pgvector database, migrations applied, and test configuration. docker compose run −−rm api pytest is the usual full backend suite; PostgreSQL-specific tests may be skipped unless that integration URL is configured, as reflected in the test fixtures/environment. CI installs the ONNX runtime extra so the ONNX provider adapter tests run instead of skipping. The full quantized model benchmark and provider/retrieval comparison were run as the M8.1 Docker proof and are not repeated on every CI run.

19. CI/CD

The only checked-in workflow is .github/workflows/ci.yml. It runs on pull requests and pushes to main, with repository contents read-only.

Backend job: Ubuntu runner starts pgvector/pgvector:pg16 as a PostgreSQL service with a health check. It sets test environment/database URL, installs Python 3.12, caches pip inputs, installs the CPU-only torch==2.8.0+cpu wheel and the project dev, local−embeddings, and deployment−onnx extras under constraints−linux−cpu.txt, then runs Ruff, alembic upgrade head, alembic check, and pytest. This exercises PyTorch and ONNX provider adapter tests in CI; the full pinned ONNX model benchmark is a separate M8.1 verification.

Frontend job: Ubuntu runner sets up Node 22 and npm cache, runs npm ci, then frontend helper tests, lint, TypeScript typecheck, and Next.js production build.

This is **continuous integration**, not deployment. It does not publish images or deploy an environment. Render auto-deploy is disabled in render.yaml; releases are manually deployed from the Render dashboard after review.

20. Failure scenarios and safe behavior

Failure	Current behavior
PostgreSQL unavailable	API database operations fail; readiness reports unhealthy. Worker cannot claim jobs and logs/retries the polling loop. No local data is invented.
Local MinIO or hosted private object storage unavailable during upload	Upload fails without a ready document; no public object URL is exposed. Hosted storage credentials remain on the backend.
Invalid, encrypted, over-limit, or over-page PDF	Upload validation rejects it; unsupported/scanned/no-text content fails processing explicitly. OCR is not attempted.
Object write succeeds, metadata/job transaction fails	Service rolls back SQL and attempts compensating object deletion. If compensation fails, it logs the opaque key for support.
Worker crashes	The job remains RUNNING until lease expiry. Another worker can reclaim it; the attempt count is bounded.
Local model download/cache fails	The local worker does not claim a document job until model setup succeeds; a transient setup failure is retried. Hosted ONNX model files are baked into the image, so there is no runtime model download; image build/deploy must succeed.
Embedding or extraction fails	Classified transient/permanent where possible; a safe error code/message and retry schedule are persisted. READY is only committed with pages/chunks/vectors.

Failure	Current behavior
Two workers see the same eligible job	PostgreSQL row locking with SKIP LOCKED allows only one claim; fencing token checks protect final writes.
Document deleted during or before work	Deletion revokes access, removes derived data and cancels the job. Worker claim/persistence also checks deletion and current lease.
User changes another workspace's ID	Membership check and workspace-scoped SQL produce 404 for inaccessible resources; composite constraints prevent cross-tenant associations.
Render backend is asleep	The frontend shows a bounded startup notice, checks readiness, and continues when the API wakes. No keep-alive request is used.
Search/Ask fails	API returns a safe structured error. UI shows a generic retry message; internal scores and exceptions are not exposed.
Weak or unsupported question	Ask does not invent an answer. It labels weak related passages as such or shows no evidence cards.

21. Architectural decisions and trade-offs

All current records are in docs/adr/. Decisions are scoped to V1; changing one requires an explicit scope/architecture discussion.

ADR	Problem → decision	Benefit and trade-off
0001 Modular monolith	Keep Next.js and modular FastAPI; worker process topology may differ by environment.	Clear domain modules without distributed-service overhead. Local API/worker are separate; constrained hosting consolidates the existing worker loop in FastAPI and shares its model session.
0002 WorkOS AuthKit	Avoid owning credentials, resets, and verification; use managed OIDC and server-side BFF cookies.	Strong identity boundary; external identity configuration and provider availability are required.
0003 Application tenant checks	Use membership authorization, mandatory scoped SQL, and composite constraints; defer PostgreSQL RLS.	Explicit and testable; every query must retain scope, while constraints provide integrity rather than read authorization.
0004 Object storage/deletion	Keep PDF bytes private in an adapter-backed object store; tombstone then clean up.	DB remains relational and access revokes immediately; object/DB cleanup needs retries because it is not one transaction.
0005 Durable worker	Use one PostgreSQL job row per document, leases, SKIP LOCKED, retries, fencing and idempotent output; no Redis/Celery.	Durable with fewer services; polling/lease behavior must be maintained and measured.
0006 Chunk/vector storage	Store one active vector on each chunk with page/hash/model metadata.	Simple retrieval join; model changes need controlled re-embedding and possibly schema changes.
0007 Hybrid retrieval	Combine Postgres FTS and pgvector ranks using configurable RRF.	Exact terms and paraphrases complement each other; RRF improves ranking but is not calibrated confidence.
0008 Product boundaries	Keep V1 to multi-user workspaces, areas, text PDFs, source evidence, no OCR/history/LLM.	Complete core journey without uncontrolled scope; users interpret evidence rather than receiving synthesized answers.

ADR	Problem → decision	Benefit and trade-off
0009 Public portfolio demo	Render Free for frontend/backend, Neon for PostgreSQL/pgvector/FTS and private S3 storage, WorkOS retained.	Public HTTPS demo at $0 recurring target; sleeping services and low quotas accepted, with no SLA. Deployment is manually triggered.
0010 Local embedding model	Keep pinned MIT intfloat /e5−small−v2, CPU, normalized 384 dimensions; PyTorch locally and ONNX INT8 for the hosted image.	Avoid paid inference and fit the 512 MiB backend limit at batch size 1. Distinct embedding-version metadata means provider changes require re-embedding.
0011 Evidence Q&A without generation	Return deterministic selected source passages; use lexical, semantic, structure, and answer-type signals.	No generative hallucination or question charge; no prose synthesis and heuristic classification can remain ambiguous.

Zero-charge portfolio constraint

The default portfolio app must not depend on usage-billed AI inference APIs. Document processing, embeddings, retrieval, and current evidence Q&A work without paid AI API calls. WorkOS remains the explicitly approved identity provider, separate from the AI inference constraint. No external paid AI provider or generative model is part of current V1.

22. End-to-end request traces

The paths below follow current request boundaries. File names are repository-relative.

A. User signs in

/sign−in route and AuthKit middleware (apps/web/src/app/sign−in/route.ts, src/proxy.ts) → WorkOS authorization/PKCE → /callback route → sealed server session cookie → server-side apiFetch() obtains short-lived token → FastAPI get_current_user() verifies JWT → resolve_authenticated_user() maps sub to User.

B. Workspace creation

apps/web/src/app/workspaces/page.tsx → createWorkspace in apps/web/src/app/actions.ts → server-only apiFetch() → backend/app/tenancy/router.py:post_workspace → tenancy/service.py:create_workspace → insert Workspace and initial OWNER WorkspaceMembership transactionally → return workspace list/redirect into workspace.

C. Knowledge Area creation

[workspaceId]/knowledge−areas/page.tsx → server action in actions . ts → POST /api/v1/workspaces/{id}/knowledge−areas → knowledge_areas/router.py checks membership and Owner role → knowledge_areas/service.py:create_knowledge_area validates normalized name and scoped uniqueness → insert row with workspace ID → return updated list.

D. PDF upload

[workspaceId]/documents/page.tsx multipart form → uploadDocument in apps/web/src/app/actions.ts → Next server calls POST /api/v1/workspaces/{id}/documents → documents/router.py checks Owner → documents/validation.py verifies structure/type/limits/page count → documents/service.py:upload_document writes via ObjectStorage → one SQL commit creates Document and ProcessingJob → UI shows Processing.

E. Worker processing and embeddings

Local: python −m app.jobs.worker → run_worker() loads the local model before claiming work → claim_next_job() locks an eligible job with PostgreSQL SKIP LOCKED, assigns lease token → process_claim() reads PDF through ObjectStorage, extracts/chunks, embeds with purpose="document" → persist_processed_output() verifies the fence and atomically persists pages/chunks/vectors and READY/SUCCEEDED. Hosted: FastAPI lifespan initializes one ONNX provider and passes that same object to Ask/search and run_worker() in a

thread; upload wakeups avoid waiting for the long idle poll. The durable claim/lease/retry/fence/commit path is unchanged.

F. Ask KnowFlow question

[workspaceId]/search/SearchForm.tsx → POST /api/workspaces/{id}/ask BFF route → server-side apiFetch() → FastAPI ask_workspace() checks membership → _retrieve_for_workspace() invokes retrieve () → query embedding + FTS + pgvector + RRF → select_evidence() extracts, assesses, ranks and deduplicates → AskResponse → UI renders evidence state and source cards.

G. Open a source PDF at a cited page

Evidence source_path points to /workspaces/{workspace}/documents/{document}?page=N → authenticated document detail page loads metadata → PDF <iframe>/viewer route uses a page fragment → Next content BFF route requests FastAPI stream → documents/router.py:read_document_content rechecks membership and active document → adapter reads private object → Next streams content to browser. A page fragment is a browser viewer hint, not an authorization mechanism.

H. Document deletion

Owner confirms delete in document UI → Server Action → DELETE /api/v1/workspaces/{id}/documents/{id} → documents/service.py:mark_document_deleted() checks Owner and workspace → commits tombstone, cancels unfinished job, deletes chunks/pages → private object cleanup → completion timestamp and purge date → no normal list/search/download returns the tombstoned document.

23. Important code to understand

Essential: recommended reading order

1. backend/app/db/models.py — tables, enums, composite constraints, vector metadata.
2. backend/app/identity/dependencies.py and backend/app/identity/workos.py — JWT trust and current-user boundary.
3. backend/app/tenancy/policies.py — membership and Owner rules.
4. backend/app/documents/router.py and backend/app/documents/service.py — authenticated resource flow.
5. backend/app/documents/storage.py and validation .py — private object abstraction and PDF boundary.
6. backend/app/jobs/service.py — durable claim, leases, retry and fence.
7. backend/app/jobs/worker.py — processing orchestration.
8. backend/app/processing/extraction.py, chunking.py, and persistence .py — source-to-chunk pipeline and atomic completion.
9. backend/app/embeddings/sentence_transformers.py — local E5 contract.
10. backend/app/search/service.py and backend/app/search/evidence.py — M5 retrieval and M6 evidence selection.
11. backend/app/search/router.py and schemas.py — API boundary and response contracts.
12. apps/web/src/lib/api.ts, apps/web/src/app/api/workspaces/[workspaceId]/ask/route.ts, and search/SearchForm.tsx — BFF-to-Ask UI.
13. apps/web/src/app/workspaces/[workspaceId]/layout.tsx and WorkspaceNavigation.tsx — current shared shell and responsive navigation.

Important

- backend/app/main.py — router mounting and error handlers.
- backend/app/db/session.py, backend/app/core/config.py — session lifecycle and validated settings.
- backend/app/tenancy/service.py — workspace/membership behavior and final Owner protection.
- backend/app/documents/lifecycle.py — tombstone purge timing.
- backend/migrations/versions/0001...0005... — actual schema history and M5 requeue.
- backend/tests/test_postgres_retrieval.py, test_postgres_job_concurrency.py — database-specific assurances.
- .github/workflows/ci.yml, docker−compose.yml, backend/Dockerfile — automation and runtime packaging.

Useful later

- apps/web/src/app/actions.ts — all server form actions and redirects.

- apps/web/src/app/workspaces/[workspaceId]/documents/[documentId]/page.tsx — source viewer path.
- apps/web/src/app/components/ConfirmSubmit.tsx, PendingSubmitButton.tsx — M7 interaction primitives.
- backend/tests/test_evidence.py, test_embeddings.py, test_processing.py — focused behavioral contracts.
- docs/adr/ — explain the design before changing it.

24. Interview preparation

Explain KnowFlow in two minutes

KnowFlow is a multi-tenant knowledge platform for internal business PDFs. WorkOS AuthKit authenticates users, while PostgreSQL workspace_memberships defines Owner/Member access. Owners organize documents into Knowledge Areas and upload private text-based PDFs. The same backend codebase claims durable PostgreSQL jobs, extracts page text with pypdf, creates deterministic page-aware chunks, and embeds them locally with a pinned CPU E5 model. Local development uses a separate worker container, PostgreSQL, and MinIO; the hosted portfolio demo uses an in-process worker and shared ONNX session, Neon PostgreSQL, and private Neon Object Storage. Search combines PostgreSQL full-text search with pgvector cosine retrieval using Reciprocal Rank Fusion. Ask KnowFlow adds a deterministic evidence-quality layer, then displays verbatim source passages and protected document/page links. It does not generate answers or call a paid AI API. This demonstrates authentication boundaries, tenant isolation, relational modeling, asynchronous jobs, failure handling, object storage, testing, and practical search engineering.

Explain KnowFlow technically in 5-10 minutes

Start with the identity boundary (AuthKit cookie in Next, server-side short-lived token, JWT/JWKS validation, immutable sub mapping). Explain that KnowFlow membership is the authorization source. Then describe the modular monolith and its two runtime topologies: local separate API/worker containers and hosted FastAPI with its in-process durable worker. Follow upload from validation through private object write and transactional metadata/job creation. Explain PostgreSQL SKIP LOCKED, leases, heartbeat and fencing token, then the pypdf/page/chunk/local embedding pipeline and atomic READY commit. Compare local PyTorch E5 with the memory-tested hosted ONNX runtime. For retrieval, contrast FTS and vector similarity, explain RRF, tenant/status/model filters, and why E5 is local. Finish with Ask's deterministic quality/deduplication/provenance layer, zero-LLM boundary, tests and trade-offs.

Questions and model answers

1. **What problem does KnowFlow solve?** It lets a workspace organize internal PDFs and locate relevant policy passages with verifiable document/page provenance, instead of relying on manual file search.
2. **Is KnowFlow an AI application?** It uses a pretrained local embedding model for semantic retrieval. It does not train a model or use a generative LLM. Its core is still a full-stack document platform and retrieval system.
3. **What is an embedding?** A numeric representation of text that lets software compare semantic similarity. In KnowFlow E5 maps a query or passage to 384 numbers; it does not produce prose.
4. **Why E5-small-v2?** It is compact, English-focused, supports retrieval prefixes, produces 384 dimensions, is MIT-licensed, and runs locally on CPU without a paid API.
5. **What does vector(384) mean?** Each database vector has exactly 384 floating-point dimensions. The model, application settings, metadata constraint, and schema must agree.
6. **What is cosine similarity?** A measure of the angle/direction between two vectors. KnowFlow orders candidates by pgvector cosine distance and derives similarity for evidence assessment; it is not a calibrated answer probability.
7. **What is pgvector?** A PostgreSQL extension and SQLAlchemy type/operator support for storing and comparing vectors. It keeps vector search in the same database as tenant/document metadata.
8. **Why hybrid search?** FTS is strong for exact terms and numbers; vector search can find paraphrases. RRF combines their rank lists without pretending their raw scores share one scale.
9. **What is PostgreSQL FTS?** Native tokenization, query matching and ranking. KnowFlow uses an English tsvector, websearch_to_tsquery, ts_rank_cd, and GIN index.
10. **What is RRF?** Reciprocal Rank Fusion adds 1/(k+rank) for each list where a passage appears. It provides a deterministic combined order; k=60 is a configurable damping constant, not confidence.
11. **Why not only vector search?** Semantically close text can be a test question, a neighboring policy, or unsupported for the requested concept. Exact keywords and evidence checks help guard against that.

12. **Why not only keyword search?** Different vocabulary such as "vacation time" versus "annual leave" can have low token overlap even when the meanings match.
13. **How can "vacation time" find "annual leave"?** E5's query and passage vectors can be near even without exact terms. FTS is complementary; the retrieved source text remains unchanged.
14. **Why no generative LLM?** V1 must run core AI functionality without usage-billed APIs and on constrained hosting. Direct evidence avoids generated paraphrase/hallucination and keeps model/runtime needs small, at the cost of no synthesized answer.
15. **How does KnowFlow prevent hallucinated answers?** It does not generate answers. It classifies source passages and labels strong, weak, or no evidence; weak related text is not presented as the best answer.
16. **How do citations work?** Extraction stores 1-based page provenance; chunks carry page ranges; M6 carries those fields into the evidence response. Protected source routing checks authorization again before streaming the PDF.
17. **What happens when evidence is weak?** The API returns weak_evidence, no best/supporting evidence, and may include subordinate related passages. no_evidence returns no cards.
18. **How do you handle conflicting policies?** Each passage keeps its own document/page provenance. KnowFlow does not synthesize a compromise or decide which source governs.
19. **How is tenant isolation enforced?** Internal membership is checked first, every data query scopes by workspace, and composite FKs prevent cross-workspace dependent rows. Inaccessible workspace resources return 404.
20. **Why not PostgreSQL RLS?** Explicit service checks and SQL scope are the approved V1 model and are directly testable. RLS would require transaction/session context and a second policy system; it may be future defense-in-depth.
21. **Why WorkOS?** It handles managed identity and verification while Next keeps its sealed HttpOnly session server-side. FastAPI verifies the short-lived JWT independently; workspace roles remain local.
22. **Why MinIO/object storage?** PDF bytes are larger binary objects; object storage handles them while PostgreSQL stores metadata and derived text. A small interface keeps MinIO replaceable.
23. **Why a PostgreSQL-backed worker queue?** Jobs must survive restarts, and PostgreSQL is already running. It avoids another service at current scale; polling and leases are explicit maintenance responsibilities.
24. **What is SKIP LOCKED?** A PostgreSQL row-lock option that makes a worker skip rows another worker has locked, so multiple workers can claim different eligible jobs without waiting on the same one.
25. **What is idempotency here?** A repeat processing attempt replaces prior page/chunk output inside one transaction rather than adding duplicates. It only commits if its lease is still current.
26. **What is a fencing token?** A per-claim random lease token checked on heartbeats and persistence. A late worker with an old token cannot commit after another worker has taken over.
27. **What if a worker crashes?** Its row remains RUNNING until the lease expires. Another worker can reclaim it; attempt limits/backoff bound retries.
28. **How would you scale this?** Measure queue age, processing duration, DB query latency and corpus size first. Add worker replicas (already supported by row locking), tune indexes and CPU/cache. Consider approximate vector indexes or a dedicated queue only when measurements justify the complexity.
29. **What would you change for enterprise deployment?** Evaluate managed PostgreSQL/object storage, backups, secret management, observability, retention/compliance requirements, rate limits across replicas, and perhaps RLS as defense-in-depth. Preserve application authorization and test tenant isolation.
30. **Why a modular monolith?** One product does not need distributed service operations. Domain modules keep code boundaries clear. Local API and worker run separately; the portfolio backend combines the worker loop only to fit the free memory budget while preserving durable jobs.
31. **How are partial failures handled?** Upload compensates object writes if SQL fails; worker commits derived data and READY together; delete commits access revocation before attempting object cleanup and retains a tombstone for retry.
32. **What tests need PostgreSQL?** pgvector cosine operators, FTS ranking/index behavior, Alembic schema and SKIP LOCKED concurrency. SQLite fixtures test much application logic but cannot validate those PostgreSQL-specific behaviors.
33. **How does E5 fit the free hosted backend?** The hosted image uses the same pinned E5-small-v2 model as portable dynamic-INT8 ONNX, CPU-only, and batch size 1. The artifact is baked into the image; one ONNX session serves Ask queries and the worker. The 512 MiB Docker proof measured 306.16 MiB process peak RSS without OOM. Local development keeps PyTorch/Sentence Transformers.
34. **Why is the hosted worker inside FastAPI?** Render Free has a tight memory budget. One backend service avoids a second process loading another model instance. It still claims durable PostgreSQL jobs with leases, retries, and fencing; it does not use in-memory background tasks. Local Compose keeps a separate worker.
35. **What happens when Render sleeps?** The frontend uses bounded readiness retries and explains that the demo is starting. After the backend starts, readiness checks PostgreSQL/pgvector and the initialized

ONNX/worker runtime. No keep-alive traffic is used. A job interrupted by shutdown remains in PostgreSQL and becomes claimable after its lease expires.

36. **Why Neon for hosted persistence?** It provides PostgreSQL compatibility including pgvector and FTS for the existing migrations and retrieval SQL, plus private S3-compatible object storage for a tiny portfolio corpus. This avoids another data abstraction; the trade-off is reliance on free-tier quotas and provider behavior.
37. **How are hosted and local embeddings kept safe?** They use the same model revision, prefixes, normalization, and vector dimensions, but separate pipeline version metadata. Paired cosine similarity was high and retrieval cases passed; the application still requires re-embedding instead of silently mixing provider-version vectors.

25. Limitations and future work

Current limitations

- Only text/selectable PDFs are supported; there is no OCR or DOCX/image ingestion.
- Ask KnowFlow does not generate, summarize, or paraphrase answers and has no persisted conversation history.
- Evidence classification is deterministic and heuristic. It can reject a valid unusual paraphrase or select a less-specific sentence when a retrieved chunk contains several related facts. For example, "How often are backups performed?" can select the quarterly restoration-testing sentence even when the same retrieved chunk also contains the daily-backup statement. This is a known M6 selection limitation observed with both embedding providers; it is not an ONNX retrieval regression.
- Hosted processing is intentionally one document job at a time and uses ONNX embedding batch size 1. Render Free can cold-start, has constrained resources/quotas, and has no SLA.
- PyTorch and ONNX vectors have separate embedding-version metadata. Do not mix the provider versions in one active corpus; re-embed documents when changing provider/version.
- The hosted object store is Neon Object Storage and remains private behind the API adapter; the portfolio demo relies on the provider's S3-compatible behavior and current free-tier terms.
- No document version comparison/change intelligence, analytics platform, invitations workflow, billing, or enterprise granular permissions are implemented in V1. Tombstone purge is opportunistic during operations rather than driven by a scheduled cleanup service.

Possible future improvements (not committed scope)

If actual users or measurements justify them, future discussions might consider OCR, version comparison, a scheduled cleanup worker, improved retrieval evaluation, a vector index for a larger corpus, stronger operational monitoring, or an optional generation feature. Generative AI and paid inference remain outside V1 and require explicit approval; none of these possibilities should be mistaken for current functionality or an approved milestone.

26. Learning roadmap

Level 1: Explain the product and architecture

Read sections 1-4 and 21. Trace the architecture diagram and be able to explain the browser, BFF, API, database, object store, and worker roles. Files: README.md, docker−compose.yml, backend/app/main.py.

Level 2: Trace identity and user requests

Read sections 6-8 and 14. Follow WorkOS sub through get_current_user, membership, a workspace-scoped router, and back through Next.js. Files: apps/web/src/proxy.ts, apps/web/src/lib/api.ts, backend/app/identity/workos.py, identity/dependencies.py, tenancy/policies.py.

Level 3: Understand relational data and asynchronous work

Read sections 5, 8-10, 15, and 17. Draw the ER diagram from memory; explain composite FKs, the lease, fencing token, retries, and atomic persistence. Files: backend/app/db/models.py, all five migrations, jobs/service.py, jobs/worker.py, processing/persistence.py, documents/service.py.

Level 4: Explain embeddings and retrieval

Read sections 11-13. Explain query/passage prefixes, vector dimensions, FTS, cosine distance, RRF, and why retrieval is not answer generation. Files: embeddings/provider.py, embeddings/sentence_transformers.py, search/service.py, search/evidence.py, search/router.py.

Level 5: Explain trade-offs and failure handling

Read sections 20-25 and ADRs 0001-0011. Practice the interview questions. For each failure, explain which transaction or retry boundary contains it and what the user sees.

Level 6: Debug and modify independently

Start Compose, apply migrations, inspect worker logs, run a focused API/processing test, and run PostgreSQL retrieval tests. For hosted maintenance, inspect Render health, Neon migration head, and the release smoke checklist in section 16. Make a small change only after locating its owning module, request contract, tenant check, migration impact, and relevant tests. Commands are in sections 16 and 18.

Appendix A. Milestone history

The main guide above describes the integrated current system, not a historical snapshot. Milestones were delivered incrementally: M1 foundation; M2 identity and tenancy; M3 Knowledge Areas/documents; M4 durable processing and chunks; M5 local embeddings and hybrid retrieval; M6 evidence-based Q&A without generation; M7 UX and application hardening; M8 free-tier Render/Neon deployment, portable ONNX embedding runtime, and release verification. M5 superseded early M4-era descriptions that discussed unpopulated vectors or no search. M7 preserved backend, retrieval, evidence, and database architecture. M8 changed hosted process topology and runtime packaging while retaining the durable jobs and approved M5/M6 behavior.